

Customer Analytics Report: RFM Segmentation & Cohort Retention Analysis

Executive Summary

This report analyzes customer purchasing behavior using the Brazilian Olist e-commerce dataset. The primary objective is to understand customer value, retention patterns, and revenue concentration using RFM (Recency, Frequency, Monetary) analysis and cohort retention analysis.

The analysis reveals that the business has strong customer acquisition but weak repeat purchase behavior. A small segment of customers contributes a disproportionately large share of revenue. Retention drops sharply after the first month, indicating an opportunity for lifecycle marketing and customer experience improvements.

Business Questions

The analysis is designed to answer the following key business questions:

- Which customers generate the most revenue?
- How frequently do customers return after their first purchase?
- Where does customer retention decline most sharply?
- Which customer segments should be prioritized for marketing efforts?
- What strategic actions can improve long-term customer value?

Dataset Overview

Source: Olist Brazilian E-Commerce Public Dataset.

The analysis combines three primary tables: customers, orders, and payments. After preprocessing and merging, the dataset represents transaction-level purchase behavior for each customer.

Key preprocessing steps included datetime conversion, filtering for delivered orders, handling missing values, and merging datasets on customer and order identifiers.

Methodology

The analysis follows a structured customer analytics workflow:

1. Data Cleaning and Preparation
2. RFM Feature Engineering
3. Customer Segmentation
4. Cohort Retention Analysis

RFM metrics were computed at the customer level. Recency measures days since last purchase, Frequency measures number of unique orders, and Monetary measures total spend. Customers were then segmented using score-based rules to enable business interpretation.

RFM Analysis Insights

Customer purchase behavior is highly skewed. The vast majority of customers make only a single purchase, resulting in an average frequency close to 1. This indicates strong acquisition but weak natural retention.

Revenue is heavily concentrated among higher-spending users. The 'Big Spenders' segment contributes the largest share of total revenue, while a significant portion of revenue also comes from 'Hibernating' customers who are no longer active.

Champions exhibit the highest monetary value and purchase frequency but represent a relatively small portion of the customer base, suggesting an opportunity to expand this segment.

Cohort Retention Analysis

Monthly cohort analysis reveals a sharp drop in retention after the first month. This pattern is typical for transactional e-commerce businesses but highlights a key growth opportunity.

Later-month retention stabilizes at a low level, indicating that customers who do not repeat early are unlikely to return organically. Early lifecycle engagement is therefore critical.

Cohort patterns suggest that improvements in onboarding, post-purchase engagement, and remarketing could materially improve customer lifetime value.

Strategic Recommendations

Based on the analysis, the following actions are recommended:

- Implement loyalty and VIP programs for Champions to protect high-value users.
- Launch targeted win-back campaigns for At Risk and Hibernating customers.
- Provide premium bundles and cross-sell offers to Big Spenders.
- Improve first-month customer experience to reduce early churn.
- Shift part of the marketing budget from pure acquisition toward retention initiatives.

These actions are expected to increase repeat purchase rate, improve customer lifetime value, and stabilize revenue growth.

Future Enhancements

Potential next steps for deeper analysis include:

- Customer Lifetime Value (CLV) modeling
- Churn prediction using machine learning
- Behavioral clustering using unsupervised learning
- Interactive dashboard development
- Cohort revenue and profitability analysis